

Vieta's Formulas

Introduction & The Quadratic Case

Roots, coefficients, and symmetric sums for degree-2 polynomials

Name: _____

Period / Section: _____

Date: _____

1. Motivation

Consider the quadratic $ax^2 + bx + c = 0$. One approach to learning about the roots is to apply the quadratic formula — but that gives us messy radicals and is slow. A far more elegant observation is this: the coefficients of the polynomial already encode certain combinations of the roots, no formula needed.

Vieta's Formulas (named after 16th-century French mathematician François Viète) make that relationship explicit. They allow us to extract the **sum** and **product** of roots directly from the coefficients.

Definition — Root

A value r is a root of the polynomial $p(x)$ if $p(r) = 0$. Every polynomial of degree n has exactly n roots (counting multiplicity) in the complex numbers, by the Fundamental Theorem of Algebra.

2. Deriving Vieta's Formulas for Quadratics

Let r_1 and r_2 be the two roots of $ax^2 + bx + c = 0$. Because r_1 and r_2 are roots, we can factor completely:

$$ax^2 + bx + c = a(x - r_1)(x - r_2)$$

Expanding the right side:

$$\begin{aligned} a(x - r_1)(x - r_2) &= a[x^2 - (r_1 + r_2)x + r_1 r_2] \\ &= ax^2 - a(r_1 + r_2)x + ar_1 r_2 \end{aligned}$$

Matching coefficients on both sides with $ax^2 + bx + c$:

Theorem — Vieta's Formulas — Quadratic Case

Let r_1 and r_2 be the roots of $ax^2 + bx + c = 0$ ($a \neq 0$). Then:

$$r_1 + r_2 = -b/a \quad (\text{sum of roots})$$

$$r_1 \cdot r_2 = c/a \quad (\text{product of roots})$$

3. Worked Examples

Example 1. Suppose p and q are the roots of $t^2 - 7t + 5 = 0$. Find $p^2 + q^2$.

Solution.

From Vieta's: $p + q = 7$ and $pq = 5$.

We use the algebraic identity:

$$\begin{aligned} p^2 + q^2 &= (p + q)^2 - 2pq \\ &= 7^2 - 2(5) \\ &= 49 - 10 = 39 \end{aligned}$$

Example 2. Let m and n be the roots of $2x^2 + 15x + 16 = 0$. Find $1/m + 1/n$.

Solution.

From Vieta's: $m + n = -15/2$ and $mn = 16/2 = 8$.

Combine the fractions over a common denominator:

$$1/m + 1/n = (m + n)/(mn) = (-15/2)/8 = -15/16$$

Example 3. The roots of $x^2 - kx + 9 = 0$ are equal. Find k .

Solution.

If the roots are equal, call them both r . Then:

$$\begin{aligned} r + r &= k \rightarrow 2r = k \\ r \cdot r &= 9 \rightarrow r^2 = 9 \rightarrow r = \pm 3 \end{aligned}$$

So $k = 2(3) = 6$ or $k = 2(-3) = -6$.

Key Tips

- When asked for a symmetric expression like $p^2 + q^2$, don't solve for the roots. Rewrite in terms of $(p+q)$ and pq .
- Common identity bank: $p^2 + q^2 = (p+q)^2 - 2pq$; $(p-q)^2 = (p+q)^2 - 4pq$; $p^3 + q^3 = (p+q)^3 - 3pq(p+q)$.
- A zero constant term means one root is 0 — don't forget this shortcut.

4. Review Problems

1. Let a and b be the roots of $x^2 - 9x + 20 = 0$. Without solving for a and b , find $a^2 + b^2$.

Work Space

Answer: _____

- 2.** Let r and s be the roots of $3x^2 - 7x + 1 = 0$. Find $1/r + 1/s$.

Work Space

Answer: _____

- 3.** The sum of the roots of $x^2 + px + 12 = 0$ is -7 . Find p .

Work Space

Answer: _____

- 4.** The product of the roots of $5x^2 - 3x + c = 0$ is 4 . Find c .

Work Space

Answer: _____

- 5.** Let p and q be roots of $x^2 - 5x + 3 = 0$. Find each expression.

- (a) $(p - q)^2$
- (b) $p^3 + q^3$
- (c) $p^2q + pq^2$

Work Space

(a) _____ (b) _____ (c) _____

5. Challenge Problems

6. [AMC Style] Let a and b be roots of $x^2 - mx + 2 = 0$. Suppose $a + 1/b$ and $b + 1/a$ are roots of $x^2 - px + q = 0$. Find q .

Work Space

Answer: _____

7. Find all quadratics $x^2 + bx + c = 0$ whose roots r, s satisfy $r + s = rs$.

Work Space

Answer: _____